Intro. to Ellip. Curves and Modular Forms
Detailed Solutions to Selected Problems

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1. Ch1 §3 Q5.

(a) If $\tilde{F}(x, y, z) \in K[x, y, z]$ is homogeneous of degree n, show that

$$x\frac{\partial \tilde{F}}{\partial x} + y\frac{\partial \tilde{F}}{\partial y} + z\frac{\partial \tilde{F}}{\partial z} = n\tilde{F}.$$

(b) If K has characteristic zero, show that a point $(x, y, z) \in \mathbb{P}^2_K$ is a non-smooth point on the curve $C\colon \tilde{F}(x, y, z) = 0$ if and only if the triple $(\partial\tilde{F}/\partial x, \partial\tilde{F}/\partial y, \partial\tilde{F}/\partial z)$ is $(0, 0, 0)$ at our particular (x, y, z). Give a counterexample if char $K \neq 0$. In what follows, suppose that char $K = 0$, e.g., $K = \mathbb{R}$.

(c) Show that the tangent line to C at a smooth point (x_0, y_0, z_0) has equation $ax + by + cz = 0$, where

$$a = \left.\frac{\partial \tilde{F}}{\partial x}\right|_{(x_0, y_0, z_0)}, \qquad b = \left.\frac{\partial \tilde{F}}{\partial y}\right|_{(x_0, y_0, z_0)}, \qquad c = \left.\frac{\partial \tilde{F}}{\partial z}\right|_{(x_0, y_0, z_0)}.$$

(d) Prove that the condition that (x, y, z) be a smooth point on C does not depend upon the choice of coordinates, i.e., it does not change if we shift to $x'y'z'$-coordinates, where $(x' \;\; y' \;\; z') = (x \;\; y \;\; z)A$ with A an invertible 3×3 matrix. For example, if more than one of the coordinates are nonzero, it makes no difference which we choose to regard as the "z-coordinate", i.e., whether we look at C in the xy-plane, the xz-plane, or the yz-plane.

(e) Prove that the condition that a given line l be tangent to C at a smooth point (x, y, z) does not depend upon the choice of coordinates.

Solution:

(a) $\widetilde{F}(x,y,z) \in K[x,y,z]$ is homogeneous of degree n, that is,

$$\widetilde{F}(\lambda x, \lambda y, \lambda z) = \lambda^n \widetilde{F}(x,y,z)$$

differentiating wrt λ, we get,

$$\frac{\partial \widetilde{F}}{\partial(\lambda x)}\frac{\mathrm{d}(\lambda x)}{\mathrm{d}\lambda} + \frac{\partial \widetilde{F}}{\partial(\lambda y)}\frac{\mathrm{d}(\lambda y)}{\mathrm{d}\lambda} + \frac{\partial \widetilde{F}}{\partial(\lambda z)}\frac{\mathrm{d}(\lambda z)}{\mathrm{d}\lambda} = n\lambda^{n-1}\widetilde{F}(x,y,z)$$

$$\implies x\frac{\partial \widetilde{F}}{\partial(\lambda x)} + y\frac{\partial \widetilde{F}}{\partial(\lambda y)} + z\frac{\partial \widetilde{F}}{\partial(\lambda z)} = n\lambda^{n-1}\widetilde{F}(x,y,z)$$

for $\lambda = 1$

$$x\frac{\partial \widetilde{F}}{\partial x} + y\frac{\partial \widetilde{F}}{\partial y} + z\frac{\partial \widetilde{F}}{\partial z} = n\widetilde{F}(x,y,z)$$

that is $x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = n\widetilde{F}$

(b) char $K = 0$. let $P = (x_0, y_0, z_0) \in \mathbb{P}^2_K$
Case (i): If $\mathbf{z_0} \neq \mathbf{0}$, so $P = (x_0, y_0, 1)$.
Since $\widetilde{F}(x,y,1) = F(x,y) \implies \widetilde{F}_x(P) = F_x(x_0,y_0)$ and $\widetilde{F}_y(P) = F_y(x_0,y_0)$
Now,
P is non-smooth $\implies F_x(x_0,y_0) = 0 = F_y(x_0,y_0) \implies \widetilde{F}_x(P) = 0 = \widetilde{F}_y(P)$
and $x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = n\widetilde{F}$ gives $x_0\widetilde{F}_x(P) + y_0\widetilde{F}_y(P) + 1\widetilde{F}_z(P) = 0$
$\implies \widetilde{F}_z(P) = -(x\widetilde{F}_x(P) + y\widetilde{F}_y(P)) = 0$
Conversely,
$\widetilde{F}_x(P) = 0$ and $\widetilde{F}_y(P) = 0$
$\implies F_x(x_0,y_0) = 0 = F_y(x_0,y_0)$, that is, $P = (x_0,y_0,1)$ is non-smooth.
Case (ii): If $\mathbf{z_0} = \mathbf{0}$ and $\mathbf{y_0} \neq \mathbf{0}$, so $P = (x_0, 1, 0)$.
Take $G(x,z) = \widetilde{F}(x,1,z) \implies \widetilde{F}_x(P) = G_x(x_0,0)$ and $\widetilde{F}_z(P) = G_z(x_0,0)$
Similar to case (i),
P is non-smooth $\implies G_x(x_0,0) = 0 = G_z(x_0,0) \implies \widetilde{F}_x(P) = 0 = \widetilde{F}_z(P)$
and $x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = n\widetilde{F}$ gives $x_0\widetilde{F}_x(P) + \widetilde{F}_y(P) = 0$
$\implies \widetilde{F}_y(P) = -x\widetilde{F}_x(P) = 0$
Conversely,
$\widetilde{F}_x(P) = 0$ and $\widetilde{F}_z(P) = 0$
$\implies G_x(x_0,0) = 0 = G_z(x_0,0)$, that is, $P = (x_0,1,0)$ is non-smooth.
Case (iii): If $\mathbf{z_0} = \mathbf{0}$ and $\mathbf{y_0} = \mathbf{0}$, so $P = (1,0,0)$.
Take $H(y,z) = \widetilde{F}(1,y,z) \implies \widetilde{F}_y(P) = H_y(0,0)$ and $\widetilde{F}_z(P) = H_z(0,0)$
Similar to case (i),
P is non-smooth $\implies H_y(0,0) = 0 = H_z(0,0) \implies \widetilde{F}_y(P) = 0 = \widetilde{F}_z(P)$
and $x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = n\widetilde{F}$ gives $1 \cdot \widetilde{F}_x(P) = 0$

Conversely,
$\widetilde{F}_y(P) = 0$ and $\widetilde{F}_z(P) = 0$
$\implies G_y(0,0) = 0 = G_z(0,0)$, that is, $P = (1,0,0)$ is non-smooth.
Counterexample if $K = \mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$

(c) Let (x,y,z) be any point on the tangent line through a smooth point $P = (x_0, y_0, z_0)$ of $C: \ \widetilde{F}(x,y,z) = 0$
Then the direction ratios $(x - x_0, y - y_0, z - z_0) \perp (\widetilde{F}_x, \widetilde{F}_y, \widetilde{F}_z)$
$\implies (x - x_0)\widetilde{F}_x + (y - y_0)\widetilde{F}_y + (z - z_0)\widetilde{F}_z = 0$
$\implies x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = x_0\widetilde{F}_x + y_0\widetilde{F}_y + z_0\widetilde{F}_z$
Using (a)
$\implies x\widetilde{F}_x + y\widetilde{F}_y + z\widetilde{F}_z = n\widetilde{F}(x_0, y_0, z_0) = 0$

(d) Given $P = (x,y,z), \ P' = (x',y',z'), \ A \in GL(3,K)$
such that $(x,y,z) = (x',y',z')A^{-1}$
$\implies \widetilde{F}(x,y,z) = \widetilde{F}((x',y',z')A^{-1})$
$\implies \widetilde{F}_x(x,y,z) = \widetilde{F}_x((x',y',z')A^{-1}) = \frac{\partial \widetilde{F}((x',y',z')A^{-1})}{\partial x}$
Using chain rule, we get $\widetilde{F}_x(x,y,z) =$
$(\because \widetilde{F}_x(x,y,z) =)$
$\implies (\widetilde{F}_x|_P, \widetilde{F}_y|_P, \widetilde{F}_z|_P) = (\widetilde{F}_{x'}|_{P'}, \widetilde{F}_{y'}|_{P'}, \widetilde{F}_{z'}|_{P'})A^{-1}$
The condition for non-smoothness of $\widetilde{F}$ at P is $(\widetilde{F}_x|_P, \widetilde{F}_y|_P, \widetilde{F}_z|_P) = 0$
$\implies (\widetilde{F}_{x'}|_{P'}, \widetilde{F}_{y'}|_{P'}, \widetilde{F}_{z'}|_{P'})A^{-1} = 0$
$\implies (\widetilde{F}_{x'}|_{P'}, \widetilde{F}_{y'}|_{P'}, \widetilde{F}_{z'}|_{P'}) = 0$ ($\because A$ is invertible)
which is the condition for non-smoothness of $\widetilde{F}$ at P'

(e) Given $P = (x_0, y_0, z_0), \ P' = (x'_0, y'_0, z'_0), \ A \in GL(3,K)$
such that $(x,y,z)A = (x',y',z')$.
Then the equation of the tangent line to $\widetilde{F}$ at P is
$x(\widetilde{F}_x|_P) + y(\widetilde{F}_y|_P) + z(\widetilde{F}_z|_P) = 0$, that is, $(\widetilde{F}_x|_P, \widetilde{F}_y|_P, \widetilde{F}_z|_P)(x,y,z)^T = 0$
But from (c), $(\widetilde{F}_x|_P, \widetilde{F}_y|_P, \widetilde{F}_z|_P) = (\widetilde{F}_{x'}|_{P'}, \widetilde{F}_{y'}|_{P'}, \widetilde{F}_{z'}|_{P'})A^{-1}$
So, $(\widetilde{F}_x|_P, \widetilde{F}_y|_P, \widetilde{F}_z|_P)(x,y,z)^T = 0$
$\implies (\widetilde{F}_{x'}|_{P'}, \widetilde{F}_{y'}|_{P'}, \widetilde{F}_{z'}|_{P'})A^{-1}(A^{-1})^T(x',y',z')^T = 0$
$\implies \cdots$
$\implies \cdots$
$\implies \cdots$

2. Ch1 §3 Q6.

(a) Let $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ be two distinct points in $\mathbb{P}^2_K$. Show that the line joining P_1 and P_2 can be given in parametrized form as $sP_1 + tP_2$, i.e., $\{(sx_1 + tx_2, sy_1 + ty_2, sz_1 + tz_2) | s, t \in K\}$. Check that this linear map takes $\mathbb{P}^1_K$ (with coordinates s, t) bijectively onto the line $\overline{P_1P_2}$ in $\mathbb{P}^2_K$. What part of the line do you get by taking $s = 1$ and letting t vary?

(b) Suppose that $K = \mathbb{R}$ or $\mathbb{C}$. If the curve $F(x, y) = 0$ in the xy-plane is smooth at $P_1 = (x_1, y_1)$ with nonvertical tangent line, then we can expand the implicit function $y = f(x)$ in a Taylor series about $x = x_1$. The linear term gives the tangent line. If we subtract off the linear term, we obtain $f(x) - y_1 - f'(x_1)(x - x_1) = a_m(x - x_1)^m + \cdots$, where $a_m \neq 0, m \geq 2$. m is called the "order of tangency". We say that (x_1, y_1) is a point of inflection if $m > 2$, i.e., $f''(x_1) = 0$. (In the case $K = \mathbb{R}$, note that we are not requiring a change in concavity with this definition, e.g., $y = x^4$ has a point of inflection at $x = 0$.) Let $P_1 = (x_1, y_1, z_1)$, $z_1 \neq 0$, and let $l = \overline{P_1P_2}$ be tangent to the curve $F(x, y) = \tilde{F}(x, y, 1)$ at the smooth point P_1. Let $P_2 = (x_2, y_2, z_2)$. Show that m is the lowest power of t that occurs in $\tilde{F}(x_1 + tx_2, y_1 + ty_2, z_1 + tz_2) \in K[t]$.

(c) Show that m does not change if we make a linear change of variables in $\mathbb{P}^2_K$. For example, suppose that y_1 and z_1 are both nonzero, and we use the xz-plane instead of the xy-plane in parts (a) and (b).

Solution:

(a) $P_1 = (x_1, y_1, z_1), P_2 = (x_2, y_2, z_2)$ are distinct points in $\mathbb{P}^2_K$ Let (x, y, z) be any point on the line $\overline{P_1P_2}$. Then $\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2} = \frac{z-z_1}{z_1-z_2} = k(say)$

But $a(x_1, y_1, z_1)$, $b(x_2, y_2, z_2)$ can also be the points P_1 and P_2 respectively,

$\implies (x, y, z) = ((ka + 1)x_1 + kbx_2, (ka + 1)y_1 + kby_2, (ka + 1)z_1 + kbz_2)$

$\implies \overline{P_1P_2} = sP_1 + tP_2$

Next,

consider the map $\phi : \mathbb{P}^1_K \to \overline{P_1P_2}$ defined by $(s, t) \mapsto (sP_1 + tP_2)$

ϕ is well defined:

$(s_2, t_2) = k(s_1, t_1)$

$\implies \phi(s_2, t_2) = s_2P_1 + t_2P_2 = ks_1P_1 + kt_1P_2 = k\phi(s_1, t_1)$

ϕ is injective:

$\phi(s_2, t_2) = k\phi(s_1, t_1)$

$\implies (s_2x_1 + t_2x_2, s_2y_1 + t_2y_2, s_2z_1 + t_2z_2) = k(s_1x_1 + t_1x_2, s_1y_1 + t_1y_2, s_1z_1 + t_1z_2)$

$\implies (s_2 - ks_1)(x_1, y_1, z_1) = (kt_1 - t_2)(x_2, y_2, z_2)$

$\implies (s_2 - ks_1) = 0 = (kt_1 - t_2)$ ($\because (x_1, y_1, z_1) \neq k(x_2, y_2, z_2)$)
$\implies (s_2 - ks_1) = 0 = (t_1 - kt_2)$
$\implies (s_2, t_2) = k(s_1, t_1)$
ϕ is surjective: (By definition)
Thus, ϕ bijectively takes $\mathbb{P}^1_K$ to $\overline{P_1P_2} \subset \mathbb{P}^2_K$.
By putting $s = 1$ and varying t we get $P_1 + tP_2$ which contains all points of $\overline{P_1P_2}$ except P_2.

(b) Solution for (b)

(c) Solution for (c)

3. Ch1 §5 Q4.

4. Show that for any fixed u, the elliptic function $\wp(z) - u$ has exactly two zeros (or a single double zero). Use the fact that $\wp'(z)$ is odd to show that the zeros of $\wp'(z)$ are precisely $\omega_1/2$, $\omega_2/2$, and $(\omega_1 + \omega_2)/2$, and that the values $e_1 = \wp(\omega_1/2)$, $e_2 = \wp(\omega_2/2)$, $e_3 = \wp((\omega_1 + \omega_2)/2)$ are the values of u for which $\wp(z) - u$ has a double zero. Why do you know that e_1, e_2, e_3 are distinct? Thus, the Weierstrass $\wp$-function gives a two-to-one map from the torus (the fundamental parallelogram Π with opposite sides glued together) to the Riemann sphere $\mathbb{C} \cup \{\infty\}$ except over the four "branch points" e_1, e_2, e_3, ∞, each of which has a single preimage in $\mathbb{C}/L$.

Solution:

$\wp(z) - u$ **has exactly** 2 **zeroes:**

Consider a fundamental domain of the form $\alpha + \Pi$, such that the boundary of $\alpha + \Pi$ misses the poles of $\wp(z)$. Then $\wp(z)$ has exactly one double pole in $\alpha + \Pi$. For any fixed u, the poles of $\wp(z)$ and $\wp(z)$ are the same. So $\wp(z) - u$ also has exactly one double pole in $\alpha + \Pi$. Thus, by Proposition 5 ($\because \wp(z) - u \in \mathscr{E}_L$), $\wp(z) - u$ has exactly 2 zeros (counting multiplicity) in $\alpha + \Pi$.

The zeroes of $\wp'(z)$ **are precisely** $\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$**:**

we have $\wp'(-z) = -\wp'(z)$ and $\wp'(l + z) = \wp'(z)$ $\forall l \in L$. Let $\alpha \notin L$, such that $2\alpha \in L$, then

$$-\wp'(\alpha) = \wp'(-\alpha) = \wp'(2\alpha - \alpha) = \wp'(\alpha) \implies \wp'(\alpha) = 0$$

Thus $\wp'(z) = 0$ for half periods, that is $\{2\alpha \in L | \alpha \notin L\}$, that is, at $\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$. Since $\wp'(z)$ has only 3 zeroes, these are precisely those.

$\wp(z) - u$ **has a double zero when** $u = e_1$ **or** e_2 **or** e_3**:**

Given $e_1 = \wp(\omega_1/2), e_2 = \wp(\omega_2/2), e_3 = \wp((\omega_1 + \omega_2)/2)$. Clearly, $\wp(z) - u$ has a double zero at $z = \alpha$ when its derivative, $\wp'(\alpha) = 0$, and $\wp(\alpha) = u$, which holds true only for $\alpha = \omega_1/2, \omega_2/2,$ or $(\omega_1 + \omega_2)/2$. Thus, $\wp(z) - u$ has a double zero if and only if $u = \wp(\omega_1/2), u = \wp(\omega_2/2),$ or $u = \wp((\omega_1 + \omega_2)/2)$

e_1, e_2, e_3 **are distinct:**

We have already proved that $\wp(z) - e_i$, $i \in \{1, 2, 3\}$ has exactly one double zero at $z = \omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$, respectively. So, if any two e_i's are equal, say to e, then the corresponding elliptic function $\wp(z) - e$ will have four zeros in Π (a double zero at each of the two corresponding half periods), contradicting the fact that $\wp(z) - e$ still has only one double pole ($4 \neq 2 \rightarrow\leftarrow$ argument principle).

4. Ch1 §6 Q1.

1. (a) Let $L = \mathbb{Z}[i]$ be the lattice of Gaussian integers. Show that $g_3(L) = 0$ but that $g_2(L)$ is a nonzero real number.

   (b) Let $L = \mathbb{Z}[\omega]$, where $\omega = \frac{1}{2}(-1 + i\sqrt{3})$, be the lattice of integers in the quadratic imaginary field $\mathbb{Q}(\sqrt{-3})$. Show that $g_2(L) = 0$ but that $g_3(L)$ is a nonzero real number.

   (c) For any nonzero complex number c, let cL denote the lattice obtained by multiplying all lattice elements by c. Show that $g_2(cL) = c^{-4}g_2(L)$, and $g_3(cL) = c^{-6}g_3(L)$.

   (d) Prove that any elliptic curve $y^2 = 4x^3 - g_2x - g_3$ with either g_2 or g_3 equal to zero, is of the form $y^2 = 4x^3 - g_2(L)x - g_3(L)$ for some lattice L. It can be shown that any elliptic curve is of that form for some lattice L. See, for example, [Whittaker & Watson 1958, §21.73]; also, we shall prove this much later as a corollary in our treatment of modular forms.

Solution:

(a) Given $L = \mathbb{Z}[i]$.
To compute $g_3(L)$, consider the bijection of L: $x \mapsto ix$:

$$\frac{1}{140}g_3(L) = \sum_{l\in L, l\neq 0} l^{-6} = \sum_{l\in L, l\neq 0} (il)^{-6} = -\sum_{l\in L, l\neq 0} l^{-6} = \frac{-1}{140}g_3(L) \implies g_3(L) = 0$$

To compute $g_2(L)$, consider the bijection of L: $x \mapsto \bar{x}$:

$$\frac{1}{60}\overline{g_2(L)} = \overline{\sum_{l\in L, l\neq 0} l^{-4}} = \sum_{l\in L, l\neq 0} \overline{(l)^{-4}} = \sum_{l\in L, l\neq 0} l^{-4} = \frac{1}{60}g_2(L) \implies g_2(L) \in \mathbb{R}$$

$g_2(L) \neq 0$:
.
.

(b) Given $L = \mathbb{Z}[\omega]$.
To compute $g_2(L)$, consider the bijection of L: $x \mapsto \omega x$:

$$\frac{1}{60}g_2(L) = \sum_{l\in L, l\neq 0} l^{-4} = \sum_{l\in L, l\neq 0} (\omega l)^{-4} = \omega^2 \sum_{l\in L, l\neq 0} l^{-4} = \frac{-1}{60}g_2(L) \implies (1+\omega^2)g_2(L) = 0$$

$\because \omega^2 \neq 1$, so $g_2(L) = 0$.

To compute $g_3(L)$, consider the bijection of L: $x \mapsto \bar{x}$:

$$\frac{1}{140}\overline{g_3(L)} = \overline{\sum_{l\in L, l\neq 0} l^{-6}} = \sum_{l\in L, l\neq 0} \overline{(l)^{-6}} = \sum_{l\in L, l\neq 0} l^{-6} = \frac{1}{140} g_3(L) \implies g_3(L) \in \mathbb{R}$$

$g_3(L) \neq 0$:
.
.

(c) Given $cL = \{cl | l \in L\}$.
Clearly,

$$g_2(cL) = 60 \sum_{l\in cL, l\neq 0} l^{-4} = 60 \sum_{l\in L, l\neq 0} (cl)^{-4} = c^{-4} \cdot 60 \sum_{l\in L, l\neq 0} (l)^{-4} = c^{-4} g_2(L)$$

and

$$g_3(cL) = 140 \sum_{l\in cL, l\neq 0} l^{-6} = 140 \sum_{l\in L, l\neq 0} (cl)^{-6} = c^{-6} \cdot 140 \sum_{l\in L, l\neq 0} (l)^{-6} = c^{-6} g_3(L)$$

(d) Given $y^2 = 4x^3 - g_2 x - g_3$.
If $g_2 = 0$ and $g_3 \neq 0$, consider the lattice $L = \mathbb{Z}[\omega]$.
Then for $c = [\frac{g_3(L)}{g_3}]^{1/6}$,

$$g_3(cL) = c^{-6} g_3(L) = \frac{g_3}{g_3(L)} g_3(L) = g_3 \text{ and } g_2(cL) = c^{-4} g_2(L) = 0$$

Thus, $c\mathbb{Z}[\omega]$ is the required lattice.
And if $g_3 = 0$ and $g_2 \neq 0$, consider the lattice $L = \mathbb{Z}[i]$.
Then for $c = [\frac{g_2(L)}{g_2}]^{1/4}$,

$$g_2(cL) = c^{-4} g_2(L) = \frac{g_2}{g_2(L)} g_2(L) = g_2 \text{ and } g_3(cL) = c^{-6} g_3(L) = 0$$

Thus, $c\mathbb{Z}[i]$ is the required lattice.

5. Ch1 §8 Q1.

   1. For the elliptic curve $y^2 = 4x^3 - g_2x - g_3$, express $\wp(Nz)$ as a rational function of $\wp(z)$ when $N = 2$.

Solution: For the elliptic curve $y^2 = f(x) = 4x^3 - g_2x - g_3$, the addition law is given by

6. Ch1 §8 Q6.

6. Show that the group $GL_2(\mathbb{Z}/2\mathbb{Z})$ is isomorphic to S_3, the group of permutations of $\{1, 2, 3\}$. For each of the following elliptic curves, describe the image in $GL_2(\mathbb{Z}/2\mathbb{Z})$ of the galois group over $\mathbb{Q}$ of the field generated by the coordinates of the points of order 2.
   (a) $y^2 = x^3 - nx$ (n not a perfect square)
   (b) $y^2 = x^3 - n^2x$
   (c) $y^2 = x^3 - n$ (n not a perfect cube)
   (d) $y^2 = x^3 - n^3$.

Solution: $GL_2(\mathbb{Z}/2\mathbb{Z})$ is generated by 2 elements, $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, which have orders 2 and 3 respectively.
Consider the map $\phi : S_3 \to GL_2(\mathbb{Z}/2\mathbb{Z})$ given by $(1\ 2) \mapsto S$, $(1\ 2\ 3) \mapsto T$). Then ϕ is an isomorphism of groups.
On the elliptic curve $y^2 = f(x)$, For (x_0, y_0) to be of order 2, $(x_0, y_0) = (x_0, -y_0) \implies y_0 = 0$.

(a) $y^2 = x^3 - nx$
$0 = x^3 - nx \implies x = 0, \pm\sqrt{n}$
And $Gal(\mathbb{Q}(\sqrt{n})/\mathbb{Q}) = C_2$, cyclic group of order 2, which is isomorphic to $\{I, S\} \leq GL_2(\mathbb{Z}/2\mathbb{Z})$.

(b) $y^2 = x^3 - n^2x$
$0 = x^3 - nx \implies x = 0, \pm n$
And $Gal(\mathbb{Q}(n)/\mathbb{Q}) = \{e\}$, the trivial group, which is isomorphic to $\{I\} \leq GL_2(\mathbb{Z}/2\mathbb{Z})$.

(c) $y^2 = x^3 - n$
$0 = x^3 - n \implies x = \sqrt[3]{n},\ \omega\sqrt[3]{n},\ \omega^2\sqrt[3]{n}$,
And $Gal(\mathbb{Q}(\omega, \sqrt[3]{n})/\mathbb{Q}) = S_3$, which is isomorphic to $GL_2(\mathbb{Z}/2\mathbb{Z})$.

(d) $y^2 = x^3 - n^3$
$0 = x^3 - n^3 \implies x = n,\ \omega n,\ \omega^2 n$,
And $Gal(\mathbb{Q}(\omega)/\mathbb{Q}) = C_3$, cyclic group of order 3, which is isomorphic to $\{I, T, T^2\} \leq GL_2(\mathbb{Z}/2\mathbb{Z})$

7. Ch1 §8 Q9. (a) and (c)

9. Each of the following points has finite order N on the given elliptic curve. In each case, find its order.

(a) $P = (0, 4)$ on $y^2 = 4x^3 + 16$
(b) $P = (2, 8)$ on $y^2 = 4x^3 + 16x$
(c) $P = (2, 3)$ on $y^2 = x^3 + 1$
(d) $P = (3, 8)$ on $y^2 = x^3 - 43x + 166$
(e) $P = (3, 12)$ on $y^2 = x^3 - 14x^2 + 81x$
(f) $P = (0, 0)$ on $y^2 + y = x^3 - x^2$
(g) $P = (1, 0)$ on $y^2 + xy + y = x^3 - x^2 - 3x + 3$.

Solution: For the curve $y^2 = f(x) = ax^3+bx^2+cx+d$ using the addition law $(x_1, y_1)+(x_2, y_2) = (x_3, y_3)$, given by

$$x_3 = -x_1 - x_2 + \frac{m^2 - b}{a} \text{ and } y_3 = -y_1 + m(x_1 - x_3), \text{ where } m = \begin{cases} \frac{y_2-y_1}{x_2-x_1} & \text{if } (x_1, y_1) \neq (x_2, y_2) \\ \frac{f'(x_1)}{2y_1} & \text{if } (x_1, y_1) \neq (x_2, y_2) \end{cases}$$

(a) $P = (0, 4)$ on $y^2 = 4x^3 + 16$
We have $(0, 4) + (0, 4) = (0, -4)$ and $(0, 4) + (0, -4) = (0, 0)$, so the order is 3.

(b) Skipped.

(c) $P = (2, 3)$ on $y^2 = x^3 + 1$
We have
$(2, 3) + (2, 3) = (0, 1)$
$(2, 3) + (0, 1) = (-1, 0)$
$(2, 3) + (-1, 0) = (0, -1)$
$(2, 3) + (0, -1) = (2, -3)$
$(2, 3) + (2, -3) = (0, 0)$ So the order is 6